



Improving urban cyclist safety and skills: Integrating a multiagent system and virtual reality training simulations

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ABSTRACT

This study investigates how integrating immersive virtual reality with a multi-agent system can improve urban cyclist training by adapting learning experiences to individual performance. Addressing the challenge of preparing cyclists for complex urban environments, the research explores whether an adaptive VR-based system can enhance hazard perception, decision-making, and compliance with traffic rules. The proposed system leverages a context-aware multi-agent framework that dynamically adjusts traffic density, environmental conditions, and scenario complexity based on user behaviour. This personalized approach ensures that training remains challenging yet accessible, fostering progressive skill acquisition in a safe, controlled simulation environment. A preliminary evaluation was conducted with eight participants over a month-long training period. Results indicated improvements in reaction times, safety distance compliance, and overall traffic rule adherence. The system's adaptability and ability to integrate into existing urban training programs suggest its potential as a scalable, data-driven tool for cyclist education.

1. Introduction

In an era characterized by increasing awareness of sustainability and efficiency in urban transportation, the bicycle has gained significant relevance [1]. This mode of transportation, valued for its low environmental impact and effectiveness in congested city traffic, has found a particularly relevant niche in the delivery services sector. However, the number of urban cyclists, especially those employed by delivery companies, has brought significant challenges regarding road safety and co-existence on the streets [2].

Traditional training methods for urban cyclists, such as theoretical lessons, printed materials, and supervised on-road practice, have significant limitations. They lack adaptability, real-time feedback, and safe exposure to high-risk scenarios, essential for effective learning. In contrast, the proposed VR-based multi-agent system provides an immersive and dynamic training environment, allowing cyclists to practice complex traffic situations safely. This approach enhances engagement and learning retention, addressing key shortcomings of conventional training programs.

Urban cycling has become a sustainable mobility alternative, yet safety remains a critical concern. Despite efforts to improve infrastructure and awareness, cycling accidents in urban environments continue to rise. According to the Spanish Directorate-General for Traffic (DGT),

in 2022 alone, there were 8,106 accidents involving cyclists, resulting in 81 fatalities, 711 hospitalizations, and 7,033 minor injuries. Compared to 2019, this represents a 3% increase in total accidents and a 10% rise in severe injuries. Additionally, broader road safety statistics indicate that in 2023, Spain recorded an average of five daily road fatalities, totalling 1,806 deaths in 101,306 accidents—marking a 3% increase from the previous year. These figures underscore the limitations of current training methods, which often fail to provide urban cyclists with adequate preparation for navigating complex traffic scenarios. This growing challenge highlights the need for more effective and scalable training solutions beyond conventional theoretical or supervised practical sessions [3,4].

This study lies at the intersection of multiple disciplines, combining principles from educational technology, urban road safety, and human-computer interaction to address the issue of cyclist safety in urban environments. From an educational technology perspective, virtual reality simulations provide immersive and personalized learning experiences. Regarding road safety, intelligent agents allow dynamic adaptation to each cyclist's needs, enhancing the acquisition of critical riding skills. Furthermore, human-computer interaction plays a crucial role in optimizing the usability and effectiveness of the training system. This study proposes an innovative solution that directly addresses

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urban cycling and sustainable mobility challenges by integrating these approaches.

Delivery cyclists face a series of unique risks. Operating in densely populated urban environments and often under time pressures to complete deliveries, these cyclists must navigate a complex and potentially dangerous urban landscape [5]. Unlike motor vehicle drivers, they are not required to have a license to operate their bicycles, which results in a wide variability in their road safety skills and knowledge. This situation is exacerbated by cyclists' general lack of proper infrastructure, such as safe bike lanes and parking spaces, forcing them to share roads with larger and often less aware vehicles [6]. Furthermore, long working hours and often inadequate equipment contribute to fatigue and stress for these workers, reducing their reaction capacity and increasing the likelihood of incidents [7].

Virtual reality (VR) emerges as a training and prevention solution in this context. Using these simulations, VR offers a safe and controlled means for bicycle users to acquire and enhance their driving skills and road safety awareness [8]. Its ability to simulate various urban scenarios, adverse weather conditions, and complex traffic situations provides a unique opportunity to prepare cyclists for real challenges in urban mobility [9].

This study aims to evaluate the efficacy of immersive technologies and multi-agent systems in improving road safety training for urban cyclists. Specifically, we focus on three primary objectives:

- **Assess the impact of adaptive VR-based training** on cyclists' ability to comply with traffic rules, improve reaction times, and maintain safe distances from other road users.
- **Develop and integrate a multi-agent system** that dynamically adjusts traffic conditions and training scenarios based on real-time user performance, providing a personalized and scalable learning experience.
- **Analyse the role of gamification and engagement strategies** in enhancing the effectiveness of virtual training, ensuring that the learning experience is interactive.

Positioned at the intersection of multi-agent systems, extended reality, and learning sciences, this study aligns with experiential and adaptive learning models in educational technology. The system fosters progressive learning and safer cycling behaviours by immersing cyclists in interactive training environments and dynamically adjusting difficulty levels based on individual performance. These insights contribute to the broader goal of integrating intelligent training solutions into urban mobility strategies, supporting sustainable and safer transport systems.

The simulation system is based on a framework incorporating intelligent agents capable of adapting and dynamically responding to user actions. This approach facilitates a personalized learning experience, adjusting training scenarios and challenges according to each cyclist's progress and specific needs. The interaction of agents within the system ensures efficient simulation management and real-time adjustment of educational interventions, essential for addressing individual variability in skills and behaviour.

Integrating VR, multi-agent systems, and road safety education represents a novel interdisciplinary approach that enhances training methodologies and transportation safety research. VR provides an immersive learning environment where cyclists can experience real-world scenarios in a controlled setting, allowing them to develop essential skills without physical risk. Meanwhile, multi-agent systems introduce dynamic and unpredictable interactions with virtual vehicles and pedestrians, simulating real-life complexities that traditional training methods cannot replicate.

While VR has been widely explored in training environments due to its ability to provide immersive and controlled simulations and multi-agent systems have been applied to enhance decision-making in dynamic environments, their combined application in cyclist training

remains largely unexplored. This study presents a novel approach by integrating a *context-aware multi-agent system* with a VR environment, enabling adaptive and personalized learning experiences beyond traditional training methods. Unlike static training programs, which provide a fixed set of scenarios, this system dynamically adjusts the complexity of training environments based on individual user performance. The multi-agent architecture allows real-time monitoring of user actions and traffic conditions, adapting scenario difficulty according to observed behaviours and common mistakes.

This synergy contributes to education research by leveraging technology to improve experiential learning. The ability to adjust training difficulty dynamically based on user performance ensures that each session is tailored to the learner's needs, fostering progressive skill acquisition. Furthermore, from a transportation safety perspective, integrating intelligent agents into VR-based training allows for systematically evaluating behavioural patterns, decision-making strategies, and risk factors, offering insights that could inform urban mobility policies and infrastructure design. Through this approach, the system promotes progressive skill acquisition, improving user engagement and retention of safety practices while maintaining a risk-free training environment.

To further highlight these advantages, Table 1 compares the proposed system and conventional cyclist training approaches, emphasizing its adaptability, safety, and engagement benefits.

As cities worldwide transition toward sustainable mobility models, cycling has become a key pillar in urban transportation strategies. While many governments promote cycling as an eco-friendly alternative to motorized vehicles, this rapid growth has exposed critical gaps in infrastructure and road safety training, particularly for last-mile delivery workers who face constant risks. Addressing these challenges, this study aligns with the global need for technological solutions that improve urban mobility safety. By integrating virtual reality simulations with multi-agent systems; this research offers an adaptable and scalable tool that enhances the training of urban cyclists, reducing accidents and fostering more sustainable urban environments.

The paper is organized as follows: Section 2 reviews related work and establishes the theoretical basis for using multi-agent systems and virtual reality in cyclist training. Section 3 details the proposed system architecture, describing how different agents interact to provide an immersive and adaptive training experience. Section 4 explains the design of the case study implemented to evaluate the system's effectiveness. In contrast, Section 5 presents the obtained results, highlighting observed driving skills and cyclist safety improvements. Finally, Section 6 discusses the implications of these results for future transport policies and road safety education. Section 7 concludes the work by summarizing the main contributions and suggesting future lines of research.

2. Related works

Technology is advancing rapidly, and Virtual Reality (VR) has emerged as a revolutionary tool in various fields, including bicycle simulators. Combining hardware and software, these simulators recreate the cycling experience in a controlled environment, opening up new possibilities in areas such as road safety education, sports training, and entertainment. The integration of VR into these systems has significantly improved the quality and effectiveness of simulation, providing immersive and real experiences that were unthinkable just a decade ago [10,11].

The evolution of bicycle simulators reflects technological advancement and the growing interest in recreational and professional cycling. These devices have evolved into sophisticated systems offering immersive and authentic experiences from the earliest rollers and stationary bikes. The integration of computerized systems in the 1990s marked a significant milestone, adding variability in resistance and terrain for a more real experience [12].

Regarding hardware, the simulators incorporate platforms that accurately simulate the vibrations and movements of cycling on different

Table 1
Comparison of the proposed system vs. Traditional cyclist training.

Aspect	Traditional Training	Proposed VR Multi-Agent System
Adaptability	Static scenarios, pre-defined routes	Dynamic scenarios, adjusts in real-time to user performance
Safety	Requires real-world exposure to traffic risks	Risk-free, controlled simulation of hazardous conditions
Engagement	Classroom-based learning, passive instruction	Immersive, interactive VR with gamification elements
Personalization	One-size-fits-all approach	Tailored difficulty and progression based on individual learning curves
Feedback	Instructor-led assessments	Real-time, automated performance feedback

terrains. These improvements have been complemented by the use of VR, which provides detailed and immersive virtual environments [13].

As for software, there has been considerable progress. Current simulators can simulate various conditions, including climatic changes and traffic dynamics, and offer the possibility to interact with other cyclists online. Additionally, artificial intelligence and machine learning are beginning to influence the customization of scenarios and challenges, adjusting to the cyclist's skills [14].

VR has transformed simulation experiences as a tool for physical training and as a medium for immersive learning and skill development. One of its most direct applications is in road safety education, where VR-based simulators provide a controlled environment for cyclists to practice navigating complex traffic scenarios without real-world risks [15]. The ability to safely experience and react to hazardous situations makes VR an essential tool in training and rehabilitation programs.

VR bicycle simulators have also gained popularity in entertainment and sports. They enable users to experience cycling routes in exotic locations or participate in virtual competitions from their homes. This recreational use has proven to be an effective incentive for promoting regular physical exercise, especially in urban settings where outdoor cycling options may be limited [16].

VR bicycle simulators also hold significant potential in rehabilitation and health. These systems can be valuable tools for physical rehabilitation, allowing patients to perform therapy exercises in a controlled and motivating environment [17]. The ability to adjust the resistance and conditions of the virtual environment facilitates a personalized approach for each patient, which is crucial for an effective and enjoyable recovery.

Physical immersion is an essential component of the experience in a VR bicycle simulator. This technical aspect focuses on how the user physically interacts with the simulator. Devices like stationary bikes, rollers, and motion platforms are fundamental for simulating the cycling experience [18]. These devices are designed to replicate the physical sensations of bicycle riding, including the resistance during pedalling and variations when traversing different terrains. The accuracy of these devices in simulating resistance and movement is crucial for a practical, immersive experience [15].

VR goggles or projection systems achieve visual immersion in VR bicycle simulators. These systems provide a detailed visual representation of the environment, which is vital for the immersion experience. The visual fidelity of the scenarios, from urban landscapes to rural routes, plays a crucial role in the credibility of the simulation experience. Auditory immersion is also an important technical aspect. The quality of sounds enhances the immersion and authenticity of the virtual environment [19].

Augmented reality (AR) is another technical facet that is beginning to be integrated into VR bicycle simulators. Unlike pure VR, which fully immerses the user in a virtual environment, AR superimposes virtual information onto the real environment [20]. In the context of bicycle simulators, this can include the display of performance metrics, directions, or interactive elements that overlay the user's visual field [21]. Integrating AR into bicycle simulators can enrich the experience by providing relevant information in an intuitive and accessible manner.

Current research in VR-assisted bicycle simulators is oriented towards continuously improving user experience and expanding their applications. A significant focus area is enhancing haptic feedback,

essential for physical immersion [22]. Advances in haptic technology are being explored to improve the correspondence between the user's physical actions and the simulator's responses [23].

A promising approach that complements haptic technology in VR-assisted bicycle simulators is the integration of multi-agent systems. These systems consist of autonomous agents interacting with each other and the user to achieve complex goals. In educational contexts and gamification through virtual reality, multi-agent systems allow for deep and adaptive customization of the environment and interactions, dynamically adjusting to the user's needs and behaviour [24].

Virtual reality (VR) and multi-agent systems (MAS) have been widely studied as practical tools for enhancing experiential learning in various educational domains. Prior research has demonstrated that VR-based training improves knowledge retention, engagement, and situational awareness in educational settings [25]. Specifically, VR simulations have been successfully applied in road safety education, allowing users to practice real-world scenarios without physical risk [26]. Multi-agent systems, on the other hand, enable personalized and adaptive learning by dynamically adjusting training content based on user performance [27]. These technologies have been proven to enhance skill acquisition, particularly in environments requiring rapid decision-making and interaction with multiple variables. By integrating these approaches, our study extends previous work and applies them to urban cyclist training, addressing an existing gap in scalable, interactive safety education.

The application of multi-agent systems through VR has become an increasingly relevant field of study. Agents within these systems can play specific roles, such as learning facilitators, opponents in simulation scenarios, or even virtual mentors [28]. This technology enables the implementation of training scenarios that continuously adapt to the user's immediate actions and their progress over time [29].

Multi-agent systems demonstrate their ability to enhance engagement and learning effectiveness significantly. Recent research has shown that these systems can facilitate richer and more motivating learning experiences, providing instant and adaptive feedback essential for developing practical and cognitive skills [30]. The adaptability and intelligence of these agents allow for customization far beyond what traditional static systems can offer [31]. The suitability of multi-agent systems to enhance feedback in educational processes in VR simulators is based on their ability to operate in highly dynamic environments and manage multiple interaction variables simultaneously [32].

The works discussed in this section highlight the importance and potential of such technologies to offer better educational experiences. Although these systems have demonstrated their ability to simulate complex traffic environments and improve road safety training, the constant evolution of technologies and demands for educational adaptability call for continuous innovations. The proposed development seeks to integrate these technological advancements more effectively, optimizing customization and dynamic response to user behaviours in real-time. This improvement focuses on enhancing the ability of simulators to provide training that not only mimics reality but also responds intuitively and educationally to user actions, thus more competently preparing cyclists to face the realities of urban traffic.

3. Proposed system

The work of delivery riders has become a fundamental part of our lives. These couriers serve as the essential link between delivery

services and us, the consumers. However, their daily tasks also entail significant risks on the streets and urban roads, where they face many challenges and hazards daily. In this context, there is an urgent need to address both the safety of these workers and the awareness of best practices for road circulation.

The main objective of the proposed system is to design and develop a software architecture and a bicycle driving simulator that provides a real experience and enables riders to learn how to navigate safely and consciously on the road. This practical training approach is crucial for reducing accidents and injuries in this labour sector.

To achieve this immersive and educational experience, virtual reality is proposed through applications designed for VR glasses. This virtual environment simulates traffic situations and everyday challenges that riders may encounter in their daily work. Interacting with these simulated scenarios allows riders to acquire practical knowledge and skills without compromising safety.

3.1. Proposed architecture

The integration of the context-aware framework into a multi-agent system is proposed to achieve the development of this system. The context-aware framework [33] allows for considering the specific context of the worker, such as their work environment, level of experience, preferences, and safety constraints. This contextual information is used to personalize safety measures and adapt protocols, training, and protective equipment to the individual needs of each user. Another advantage is that the context-aware framework enables continuous monitoring of the user's progress and dynamic adjustment of preventive measures and road safety education programs. As the worker progresses in training, the system can update goals and approaches to maintain adequate safety and motivation in the work environment. Integrating this framework into a multi-agent system provides a powerful and flexible solution. Agents can interact and collaborate in real-time, sharing contextual information and adapting their decisions and actions according to the context of the environment. This allows for further personalization of safety measures, adjustment of protocols and procedures according to the situation and risks at the workplace, and enhancement of the efficacy and efficiency of safety measures in real-time.

The proposed architecture stands out for its ability to incorporate new functionalities and adapt to the environment. To achieve optimal operation, the architecture must have well-defined features. For this purpose, virtual organizations of security agents are employed, where agents have specific roles, responsibilities, and established relationships with each other. These organizations allow them to share information, coordinate actions, negotiate agreements, and make collective decisions to achieve shared goals in occupational safety. The virtual organization provides a framework to regulate and facilitate the interaction among security agents, promoting cooperation and efficiency in the occupational safety management system.

In the architecture (Fig. 1), the modules and agents are designed to accomplish specific tasks related to user training. One significant benefit of this setup is the flexibility to replace individual agents or groups of agents with similar functions without disrupting the overall system. This specialization of modules and agents enhances the management of risks and safety measures, allowing the system to adapt to evolving environmental needs. The fundamental virtual organizations of agents within this architecture are detailed below.

Data Gathering Organization: This organization collects user data, including their physical state and actions in the real world, for analysis and transfer to a virtual setting. The agents within this organization include:

- **Movement Sensor Agent (MSA):** Collects data from motion sensors on a simulated bicycle, including metrics like speed, incline, and direction.

- **Resistance Simulation Agent (RSA):** Interfaces with the roller to simulate and log riding resistance and force exerted by the virtual cyclist.
- **Health Monitoring Agent (HMA):** Tracks physical metrics such as heart and blood oxygen levels to simulate real conditions.
- **Data Transmission Agent (DTA):** Acts as a mediator between the data gathering agents (MSA, RSA, HMA) and the system, aggregating sensor data and forwarding it to the Context Analysis Organization (CAO).

Context Analysis Organization: This organization handles system deployment and modules for data feature extraction and classification. The basic agents here are adaptable for specific case studies but generally include:

- **Data Preprocessing Agent (DPA):** Acts as a preliminary step for feature extraction and context management, consolidating and filtering data from the gathering organization.
- **Feature Extraction Agent (FEA):** Analyses raw data to identify user and environmental characteristics.
- **Context Classification Agent (CCA):** Processes extracted features to produce mid-level information by combining data points to identify key details.

Application Integration Organization: This group adapts system-generated data for the application layer, serving as an interface for applications to interact with the system. The Application Handler Agent (AHA) facilitates these communications, converting raw system data into user-friendly formats. It includes:

- **Virtual Environment Agent (VEA):** Displays the virtual environment and traffic scenarios for the driving simulation.
- **Web Application Agent (WAA):** Monitors and interacts with the driving simulation, providing an interface for driving experts.

System Monitoring Organization: Manages and controls the platform's virtual organizations and agents and the context data they collect. The agents in this organization include:

- **Service Integration Agent (SIA):** Facilitates communication between external services and the virtual agent organization, managing high-level context analysis and global system information distribution.
- **System Monitor Agent (SMA):** Regularly checks the system's status, identifying information overloads, agent conflicts, resource failures, and state inconsistencies.
- **Administration Agent (AA):** Lists all available services agents can request. New agents must register their services with this agent to inform others about available services.
- **Timestamp Agent (TA):** Adds date and time information to data before storage.
- **Context Reasoning Agent (CRA):** Generates high-level context from the output of the context analysis organization, identifying patterns, irregularities, and user needs.
- **Data Management Agent (DMA):** Manages all data generated by the system, acting as a database interface and ensuring privacy and security.

Adaptation and Progress Organization: This organization adapts user data and actions to improve training progress and motivation. The agents in this organization include:

- **Adaptive Motivation Agent (AMA):** Analyses user context to adapt to drive challenges and maintain high motivation. Its functions include:
 - Analysing user training history, preferences, and progress.
 - Adjusting exercise intensity and difficulty to fit user needs.

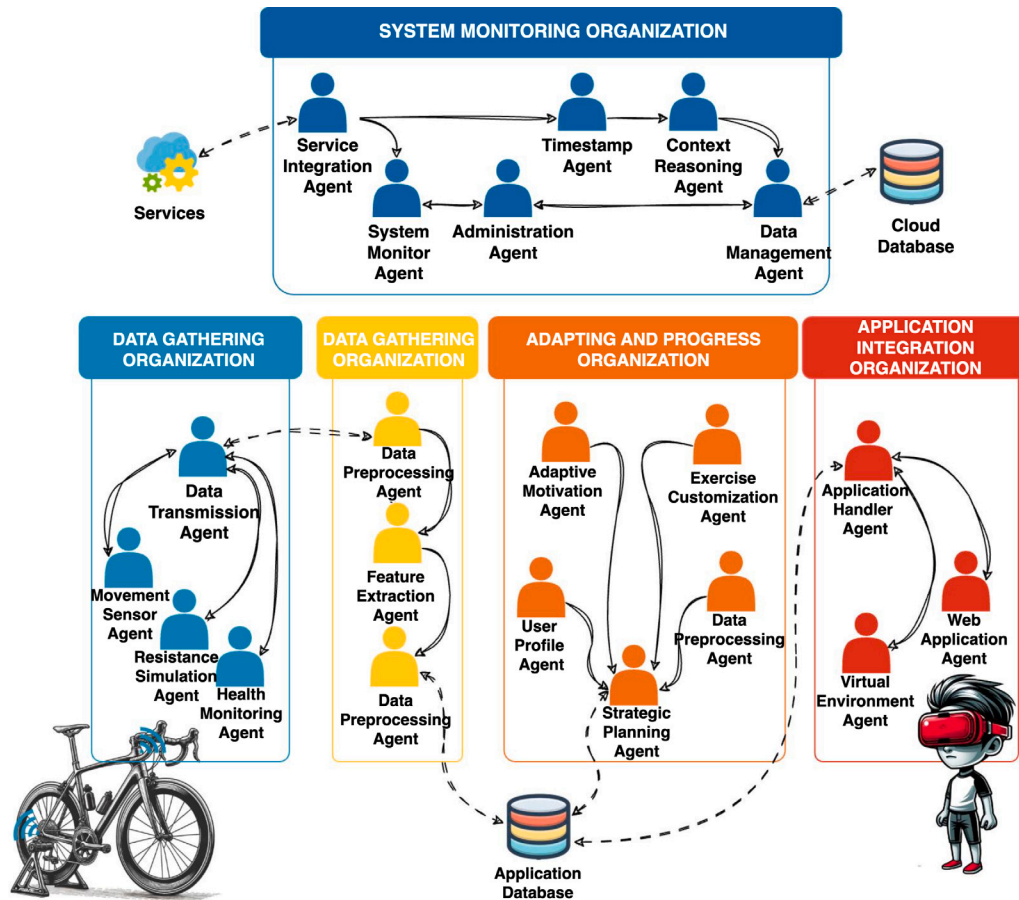


Fig. 1. System architecture.

- Designing challenges and rewards to keep the user motivated.
- Providing positive feedback to encourage overcoming obstacles.
- Progress Tracking Agent (PTA): Monitors user progress and adjusts exercises based on performance. Its functions include:
 - Evaluating performance in each exercise.
 - Comparing current results with previous records and goals.
 - Identifying areas for improvement and adjusting exercises.
 - Recording and visualizing progress over time.
- Exercise Customization Agent (ECA): Adapts exercises based on user context, including:
 - Adjusting resistance, speed, and other parameters based on sensor data.
 - Modifying exercise duration and intensity according to physical state.
 - Offering personalized exercise options aligned with user goals.
 - Updating challenges to maintain appropriate difficulty.
- User Profile Agent (UPA): Shares user information with other agents, updating profile values as progress is made.
- Strategic Planning Agent (SPA): Makes strategic decisions about exercises and challenges, including:
 - Using context and data from other agents to plan the next steps.
 - Adjusting exercises based on user response and progress.
 - Modifying activity focus to increase motivation.
 - Planning future exercises to foster continuous improvement.

Regarding agent communication, the system uses MQTT (Message Queuing Telemetry Transport) [34]. This lightweight messaging protocol is designed for minimal bandwidth and device resource usage, ensuring efficient and real-time communication. Its various broker implementations and libraries facilitate communication between agents in different environments, making it ideal for this platform. The system employs Eclipse Mosquitto as the MQTT broker, ensuring seamless message delivery between agents.

The system uses the Java Agent Development Framework (JADE) [35] to manage multi-agent interactions and decision-making processes. JADE provides an efficient structure for agent-based communication, task delegation, and context-aware processing. It enables dynamic scenario adaptation based on real-time sensor data and user interactions, ensuring a responsive and personalized training experience.

The system employs multiple databases to ensure proper functionality and scalability. The monitoring organization's database stores information about system agents, their services, and tasks. It also keeps user context information determined by the organization's agents. Additionally, a separate database stores case study-specific information. Data logging and performance tracking are managed using SQLite, which ensures lightweight and efficient data storage.

Fig. 2 shows how user sensor data is processed by different agents, translated into virtual movements, and evaluated within the simulated environment to determine warnings and penalties.

3.2. Simulator platform

The simulation structure designed for this study is a set of components crafted to create a real and immersive urban cycling experience. At the core of the simulator platform is a synergistic interaction between hardware and software, engineered to replicate the various

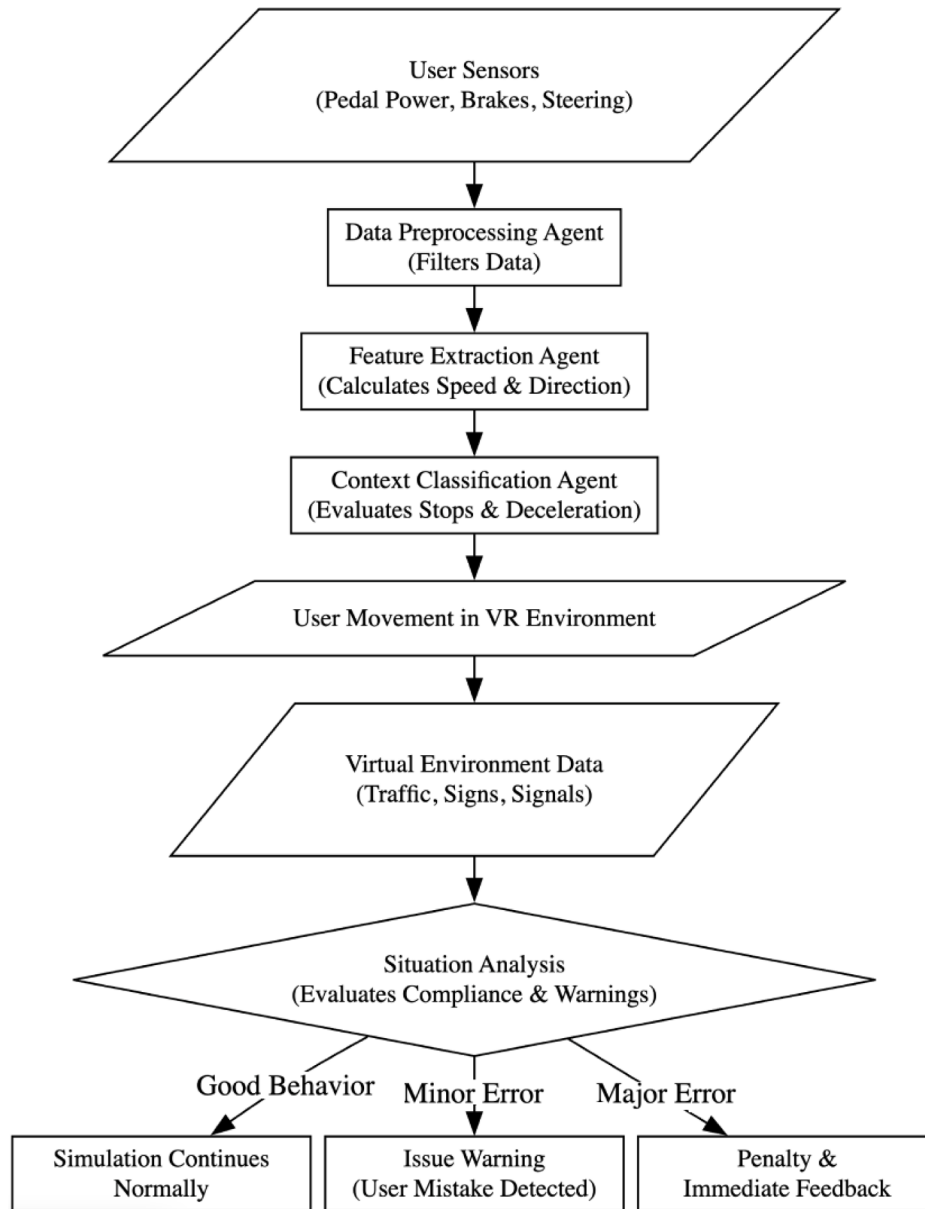


Fig. 2. Workflow of the multi-agent system.

movements and visual cues inherent to urban cycling. The developed platform can be observed in Fig. 3.

The platform simulator is anchored to a reinforced metal base to replicate urban cycling conditions. The Tacx FLUX S Smart [36] smart trainer is used for resistance and slope simulation, allowing real-time resistance adjustments through electronic control. This trainer is compatible with Bluetooth and ANT+, facilitating seamless integration with the simulator and enabling precise power, speed, and cadence measurements. Additionally, a chest-strap heart rate monitor is incorporated into the system to track the cyclist's physiological response during training. This monitor utilizes the Bluetooth and ANT+ protocols, ensuring compatibility with the simulation software and enabling real-time heart rate data transmission, which can be used for adaptive training adjustments and performance analysis.

To simulate inclines and terrain changes, the platform incorporates a 12 V linear actuator with a 25 cm stroke, capable of adjusting the incline angle from -5° to $+20^\circ$. This actuator is mounted on a high-strength ball joint system, allowing smooth inclination transitions without excessive vibrations. Its control is managed by an IBT-2

(BTS7960) motor controller, a dual-channel H-bridge module capable of handling up to 43 A of current, precisely regulating the actuator's speed and direction. This controller is connected to an ESP32-WROOM-32D microcontroller, responsible for receiving and processing real-time data from an Adafruit BNO055 Inertial Measurement Unit (IMU) [37] mounted on the bicycle's handlebars. The IMU provides tilt and rotation measurements at a 100 Hz sampling rate and features an integrated sensor fusion processor, enabling accurate orientation data without requiring additional calculations from the microcontroller.

The Adafruit BNO055 IMU measures the cyclist's tilt on the platform and detects real-time handlebar rotation, allowing the simulation of turns within the virtual environment. Thanks to its ability to measure angular acceleration and absolute orientation, the system can precisely interpret the cyclist's steering angle, ensuring a smooth and natural riding experience.

For simulation visualization, the system offers two options:

- **Traditional Screen Mode:** The user can view the virtual environment on a TV or monitor, providing an accessible training experience without requiring a virtual reality headset.



Fig. 3. Proposed Simulator Platform.

- **Immersive Virtual Reality (VR) Mode:** For a fully immersive experience, an Oculus Quest 1 VR headset allows the cyclist to interact with the virtual environment in first-person view. Although this model was used initially, the system is compatible with any other VR headset that supports PC connections, including Oculus Quest 2, Meta Quest 3, and HTC Vive. The Oculus Quest headset connects via Oculus Link, enabling real-time simulation streaming with minimal latency.

Before each session, the system is calibrated to ensure accurate synchronization with the virtual environment. The IMU, mounted on the handlebars, is calibrated statically and dynamically. Initially, the bicycle is placed on a level surface to establish a neutral reference point, ensuring accurate baseline tilt values. The user then performs controlled tilting and steering movements, allowing the IMU to fine-tune its internal sensor fusion algorithms and optimize the inclination and handlebar rotation detection. Meanwhile, the actuator responsible for platform inclination adjustments is calibrated by moving it to its extreme positions to define mechanical limits. Intermediate angles are then tested to ensure smooth and precise transitions. The motor controller, managed by the ESP32, regulates the actuator's movement using PWM signals and a PID control system to ensure responsiveness without overshooting. Once calibrated, real-time tests verify synchronization between the cyclist's movements and the simulation, ensuring a seamless and immersive experience.

The software interface is designed to communicate wirelessly with the platform hardware, receiving input from sensors and actuators to maintain a consistent experience. For example, as the cyclist approaches a virtual incline, the platform responds with a corresponding physical adjustment, simulating uphill effort. Conversely, a descending slope is mimicked by decreasing resistance, providing a sensation of gliding.

3.3. Adaptive training logic

To adapt the training difficulty dynamically, the system employs an ontology-driven approach to structure knowledge about user performance and training scenarios. Additionally, Bayesian networks model the probabilistic relationships between performance metrics and difficulty adjustments.

The system integrates multiple sensors to capture real-time performance metrics and ensure accurate evaluation of user behaviour

as indicated in Section 3.2. To ensure accurate data processing, the following techniques are applied:

- **Preprocessing:** Raw sensor data undergoes filtering using a Kalman filter to minimize noise and improve signal stability.
- **Calibration:** Before each session, sensor values are calibrated to account for individual differences and avoid measurement bias.
- **Synchronization:** All data streams (sensor inputs, virtual environment states, and user actions) are synchronized in real time using a time-stamped event queue to prevent latency issues.

Furthermore, the system monitors user performance during each session through key metrics:

- **Traffic Errors:** Ignoring signals or failing to stop at intersections.
- **Collisions:** Incidents involving pedestrians or vehicles.
- **Reaction Times:** Delays in responding to critical stimuli.
- **Safety Distance Compliance:** Consistency in maintaining safe distances from other entities.

The combination of ontologies and Bayesian networks is particularly suited for this application due to their complementary strengths in handling structured knowledge and probabilistic decision-making. Ontologies provide a well-defined semantic framework representing relationships between user actions, environmental factors, and training outcomes, ensuring a standardized interpretation of training scenarios. On the other hand, Bayesian networks allow for adaptive learning by quantifying uncertainties in user performance and dynamically adjusting difficulty levels based on observed trends.

These metrics are integrated into an ontology that defines the relationships between:

- User actions (e.g., braking, steering, acceleration).
- Environmental factors (e.g., traffic density, weather conditions, signage).
- Training outcomes (e.g., skill improvement, error reduction).

Moreover, a Bayesian network models the probabilistic relationships between user performance metrics and training difficulty. The network includes:

- **Nodes:** Represent performance metrics (e.g., error rate, reaction time, safety distance) and difficulty levels (e.g., increase, maintain, reduce).

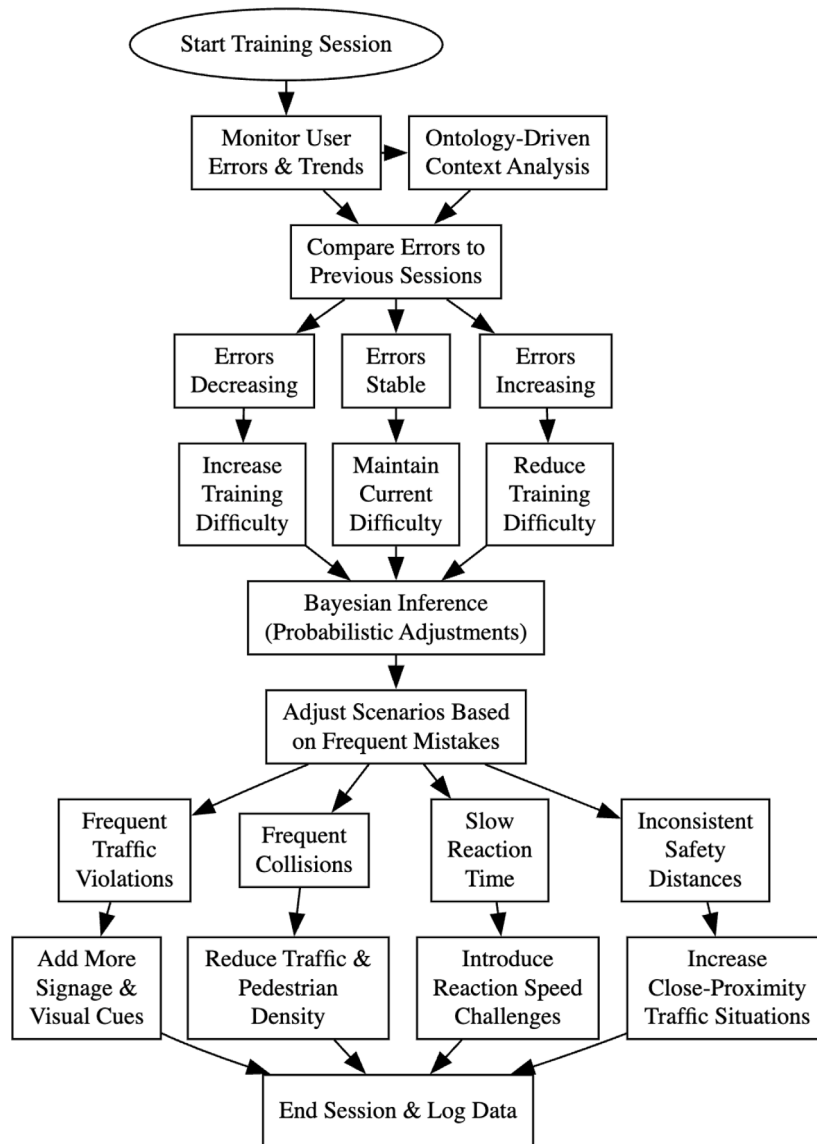


Fig. 4. Decision tree illustrating how difficulty levels are adjusted based on user performance trends, using an ontology-driven framework and Bayesian inference.

- **Edges:** Represent probabilistic dependencies, such as how an increased error rate affects the likelihood of reducing difficulty.

Based on observed performance, the Bayesian network performs probabilistic inference to determine the most appropriate difficulty level:

- The system increases difficulty if the probability of consistent improvements (e.g., error rate reduction higher than 20%) is high.
- The system reduces difficulty if the probability of frequent violations (e.g., traffic errors higher than 70%) is high.
- If performance is stable, the difficulty remains unchanged.

Fig. 4 illustrates the decision-making process for difficulty adjustment, where the ontology informs the Bayesian network, and the output adjusts the training scenarios.

This approach ensures consistent interpretation of user behaviour across different scenarios. Also, it handles uncertainty in user performance metrics and allows for adding new metrics or scenarios without modifying the underlying structure.

3.4. Gamification and transfer to virtual environment

The concept of gamification is important for transferring real-world cycling techniques and experiences to the virtual environment. Employing game design elements in non-gaming contexts transforms urban cycling into a more engaging and educational experience. This approach encourages participation, enhances motivation, and promotes effective learning of safe and responsible cycling behaviours.

The software developed in Unity incorporates typical game elements such as points, levels, challenges, and rewards to realize this concept. Points are awarded for obeying traffic rules, maintaining safety distances, and quick reaction times during emergencies. The scoring system dynamically adjusts the weight of points based on the complexity of the task, emphasizing higher rewards for more challenging scenarios. Difficulty levels dynamically adjust to the cyclist's skill level, offering incremental challenges that simulate complex urban scenarios and varied traffic situations. Rewards include unlocking new levels or challenges and cumulative scores that allow users to compare their performance with their peers anonymously.

Users can experience the simulation from a first-person perspective, enhancing immersion and realism. Including moving traffic and pedestrians enhances the simulator's authenticity, preparing cyclists for



Fig. 5. Game modes.

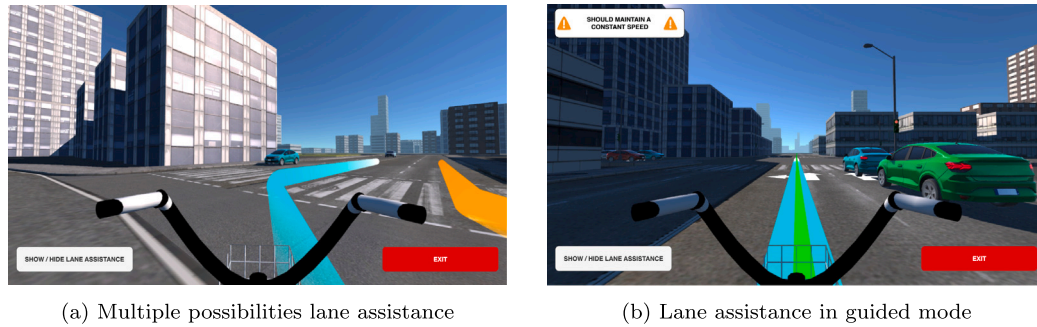


Fig. 6. Lane assistance examples.

real-world road conditions. The virtual city's streets and avenues are replicas of their real-life counterparts, with high-resolution textures and detailed models.

The simulator software offers two modes: an unguided mode and a guided mode, as seen in Fig. 5. In the unguided mode, users navigate freely without directions. In contrast, the guided mode provides a visible line on the virtual path, indicating the recommended route and facilitating learning and navigation in complex situations.

The assisted mode in the simulator software is designed to provide detailed guidance during user navigation. When multiple routes are possible, this mode displays all available alternatives, allowing the user to choose based on preference or necessity. The system distinctly indicates this sole direction in scenarios with only one viable route. This type of assistance ensures that users can make informed decisions while becoming accustomed to the dynamics and challenges of simulated traffic. Fig. 6 shows an example of this.

One of the key objectives of gamification in this context is to facilitate the transfer of skills learned in the virtual environment to real-life situations. By simulating urban cycling experiences and presenting virtual consequences for decisions made, users can practice and improve their reactions and decisions in a controlled and safe environment. This is achieved without the risks associated with training on actual streets and with the advantage of repeating specific situations as often as necessary for learning. Participants receive real-time visual and auditory feedback for their performance. For example, incorrect lane changes or violations of traffic rules triggered visual warnings (e.g., red overlays with text explanations) and auditory cues (e.g., beeping sounds). The consequences manifest in different warnings and errors displayed to the user to correct and learn from their actions in real-time, as seen in Fig. 7.

For example, as shown in Fig. 7(a), if a user changes lanes incorrectly, they will receive a clear visual warning. Other errors include deviating from the set route, collisions with other vehicles, driving in the opposite direction, and traffic signal violations, such as red lights. Each error or warning has an icon and an explanatory message to facilitate instant understanding of what went wrong and how to

correct it, as shown in Figs. 7(b) to 7(e). This allows users to recognize and adjust their behaviours in a simulated environment before facing similar situations in the real world.

Combining an interactive physical environment with a gamified virtual environment provides a powerful platform for training and evaluating urban cyclists. Users not only enjoy a captivating simulation experience but also benefit from a unique opportunity to improve their cycling skills and behaviours. The seamless transition of what is learned in virtuality to reality has the potential to significantly increase road safety and awareness of cyclists in urban environments.

A key aspect of this training system is its ability to simulate dynamic urban cycling conditions. Unlike static training programs, the proposed system dynamically adjusts the virtual environment to mirror real-world challenges, reinforcing decision-making and hazard perception skills in complex traffic scenarios. The VR environment is designed to replicate the unpredictability of urban cycling, incorporating elements such as:

- **Traffic Complexity:** Real-time traffic flow with variable vehicle densities, pedestrian crossings, and dynamic traffic signal timing based on real-world behaviours.
- **Environmental Conditions:** Diverse weather conditions affect road surface traction and visibility, including rain, fog, and night-time visibility constraints.
- **Road Infrastructure:** Detailed urban environments with intersections, roundabouts, bicycle lanes, and road signs accurately modelled from real-world locations.

The training's adaptive nature ensures that users encounter a wide range of traffic conditions and road challenges, progressively adjusting based on their performance. This enhances the transferability of skills from the virtual environment to real-world urban settings.

Additionally, the system incorporates randomized pedestrian behaviours, unexpected vehicle manoeuvres, and simulated roadwork zones, adding variability to the training experience. These unpredictable elements increase user engagement while reinforcing hazard anticipation and adaptive decision-making. By balancing realism and

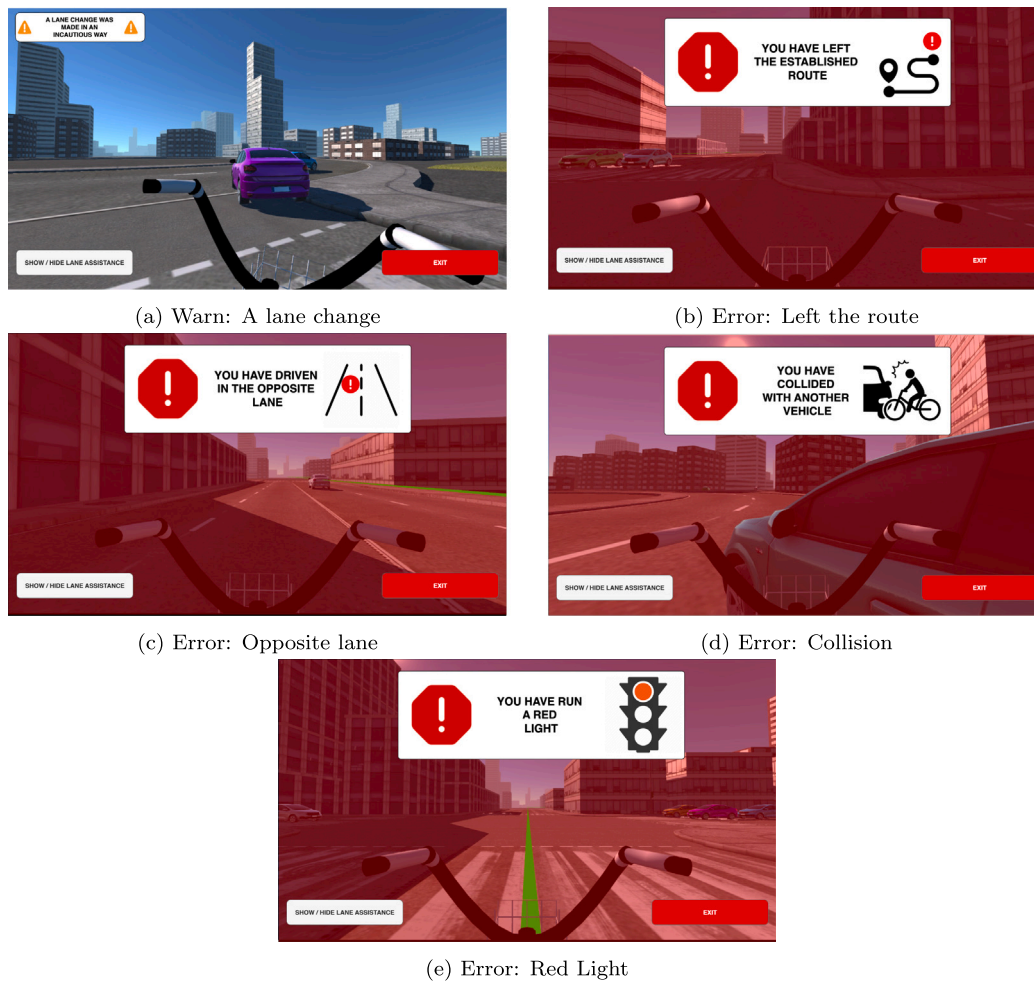


Fig. 7. Warnings and errors.

challenge progression, the VR environment ensures cyclists develop practical safety skills applicable to real-life urban mobility.

4. Case study design

To assess the system, a competency evaluation in the context of bicycle driver training represents a crucial component to ensure the educational process's effectiveness and improve road safety in urban environments. For this, measuring the users' ability to operate safely and comply with traffic regulations is important, as well as providing an objective and detailed assessment of their driving skills. Fig. 8 shows a relation between the initial objectives, the employed technologies and the expected outcomes to certify that the system gets the objectives.

A rigorous evaluation system is justified by promoting safe practices and road awareness, thus reducing the risks of using personal mobility vehicles. Among the metrics provided by the system and to be evaluated are:

- **Traffic Signal Obedience Rate:** Evaluating the user's ability to recognize and obey traffic signals is essential for road safety. A high obedience rate indicates an adequate understanding of the regulations, reducing the risk of accidents and improving safe integration into urban traffic.
- **Reaction Time in Emergency Situations:** Measuring the time it takes a user to react to emergencies reflects their ability to make quick and safe decisions. A shorter reaction time is associated with a greater ability to avoid potential accidents and mitigate risks.

- **Compliance with Safety Distances:** Measuring whether users maintain safe distances from other vehicles and pedestrians reflects their awareness and respect for the space of other traffic participants. Good compliance with distances contributes to the prevention of collisions and the maintenance of a safe environment.
- **Distance Travelled without Incidents:** Measuring the total distance travelled by the user without incidents, such as collisions or violations of regulations, reflects their ability to maintain safe behaviour and prevent dangerous situations.

A case study has been implemented to comprehensively evaluate the competencies described earlier. The main objective of this study is to analyse the effectiveness and safety of the training platform. For this purpose, a group of 8 users was selected to participate in the study. Participants were aged between 22 and 31 years (mean age: 27.5). All had previous cycling experience. Familiarity with VR was assessed before the study, revealing 2 participants with no experience and 6 with basic knowledge. These variables were not randomized but considered in the post-analysis to evaluate their influence on the outcomes. Although this sample size may be relatively small, it was strategically chosen to allow detailed evaluation and close observation of each participant throughout the sessions. Additionally, the aim was to represent a diversity of user profiles, including both novice drivers and those with experience.

Each user participated in 12 sessions, spread over a month, with 3 sessions per week. Distributing the sessions over a month introduces variability in the simulated driving conditions. Each training

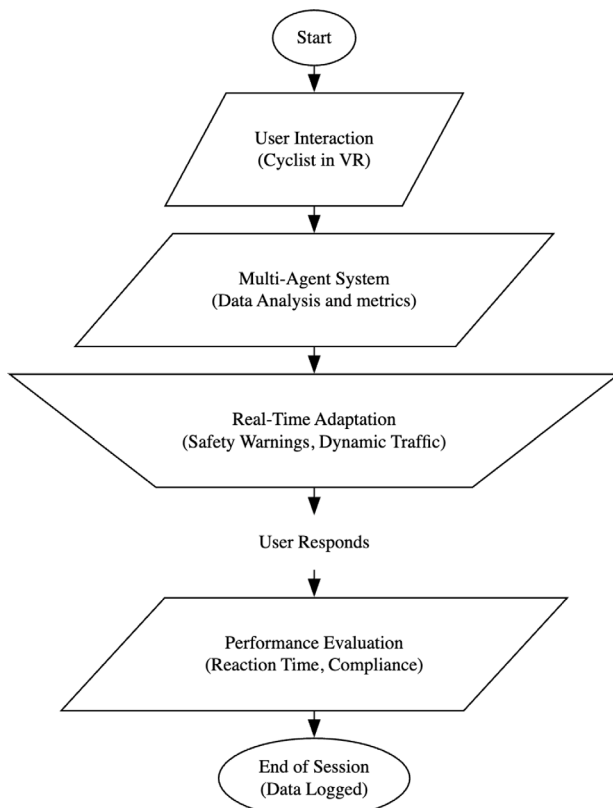


Fig. 8. Relationship between study objectives, employed technologies, and expected outcomes.

session included diverse traffic conditions, such as low (35% of scenarios), medium (35%), and high-density traffic (30%). Weather conditions were simulated to reflect real-world variability, including sunny, rainy, and foggy environments. Participants encountered specific challenges such as navigating crosswalks with unpredictable pedestrians and managing safety distances around other vehicles. This variability contributes to a more comprehensive evaluation of driving skills, ensuring that users are exposed to a wide range of situations they might encounter in real life.

Each training session lasted one hour and was designed to simulate real-world conditions progressively. Sessions began with a 10 min warm-up in scenarios featuring low traffic flow, allowing participants to familiarize themselves with the VR environment and the controls. Following the warm-up, traffic density and scenario complexity increased incrementally to challenge participants' decision-making and adaptability.

Tasks during the core 50 min training were randomized across sessions to prevent predictability and mimic real-world variability. These tasks included reacting to emergencies, maintaining safety distances, and complying with traffic signals. Randomization ensured participants were exposed to diverse situations, enhancing the validity of the training.

The study design allows detailed tracking of each user's progression throughout the 12 sessions. This length and frequency of the study enable a longitudinal evaluation, identifying improvement patterns, areas of challenge, and adaptations in user behaviour as they progress in training. Fig. 9 represents the interaction between the cyclist, the multi-agent environment, real-time adaptation mechanisms, and the performance evaluation process.

5. Results

The present study evaluated the impact of a virtual reality training program aimed at urban cyclists to improve crucial aspects related to road safety, as described in Section 4.

The results have been quantified and presented to provide a clear view of the users' evolution and progress throughout the training program. This way, the results for each established metric will be analysed. A statistical analysis of the results will also be provided to verify their truthfulness and reproducibility.

5.1. Metrics results

OBEEDIENCE RATE

The results obtained for this metric can be seen in Fig. 10. Several key observations and conclusions can be highlighted by analysing the provided graph, which displays the obedience rate in each session for a set of users during tests in a virtual reality environment. The overall trend shows an improvement in the obedience rate from the first to the last session. This suggests that the accumulated experience in the virtual environment effectively increases users' compliance with traffic signals and driving rules. The dashed line representing the average obedience rate of all users increases from the first to the last session, indicating steady and substantial progress in user behaviour.

Fluctuations are observed in the intermediate sessions for some users, which could indicate variations in the difficulty of the presented traffic scenarios or the influence of external factors on individual learning. These variations are expected to provide valuable information for adjusting the training system to make it more adaptive to each user's learning needs.

One of the most positive aspects that can be inferred from the data is the ability of all users to improve and adapt to the virtual environment over time. Even those who started with lower obedience rates managed to approach the group average in the final sessions, demonstrating the effectiveness of the simulator in developing safe driving skills.

REACTION TIME

The results obtained for this metric can be seen in Fig. 11. Analysing the provided graph that shows users' reaction time in risk situations during various virtual reality training sessions, patterns and trends that reflect the impact of the simulator on improving user skills can be identified. A notable decrease in the average reaction time of all users from the first to the last session indicates an improvement in the ability to respond quickly to risky situations. The dashed line representing the overall average decreases from more than 700 ms initially to around 400 ms in later sessions. This improvement reflects greater familiarity with risk scenarios and an enhanced ability to anticipate and react to unexpected situations.

Considerable variability is observed among users in each session, which could indicate differences in learning speed and adaptation to the simulator. Some users achieve consistently low reaction times, while others show more significant fluctuation, which could indicate the need for customized training adjustments.

Users seem to adapt to the virtual environment progressively, as demonstrated by the continuous improvement in their reaction time. This could be interpreted as a combination of increased competence in handling the virtual bicycle and heightened sensitivity to visual and auditory stimuli signalling imminent risk situations.

The decreasing trend in reaction time suggests that the simulation environment effectively teaches and reinforces quick and safe decision-making. This is particularly relevant for real-life traffic situations where cyclists must respond immediately to avoid accidents.

SAFETY DISTANCE

The results obtained for this metric can be seen in Fig. 12. The provided graph reflects the compliance with the average safety distance

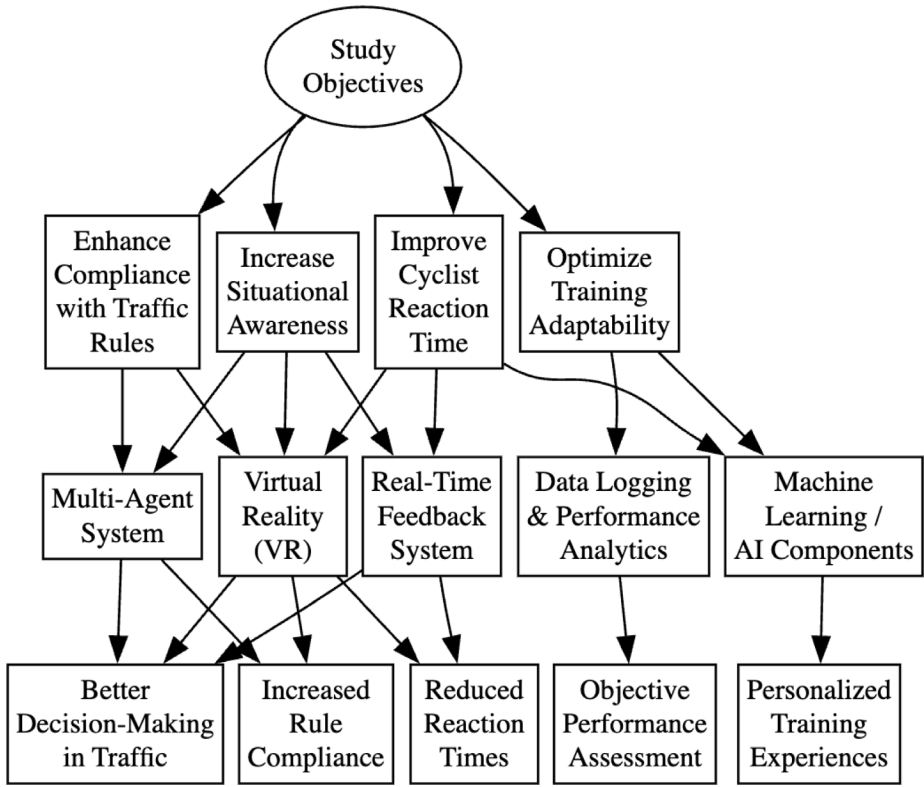


Fig. 9. Workflow of the VR-based cyclist training system.

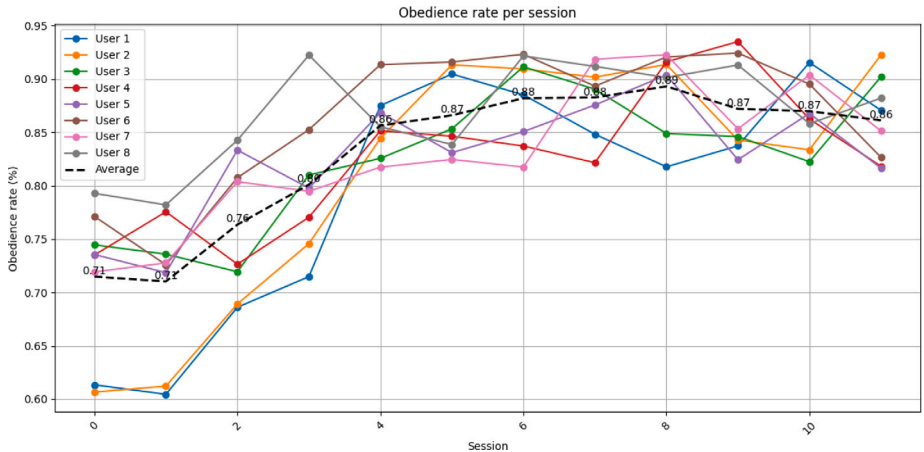


Fig. 10. Obedience Rate Results. The results show a progressive improvement across training sessions, with a notable increase in compliance observed after the second session. While early sessions exhibit more variability, later sessions demonstrate more stable adherence to traffic rules, suggesting an effective learning curve.

maintained by each user in successive virtual reality training sessions. A significant increase in the average safety distance the users maintain from the beginning to the end of the training period is observed. This suggests that users learn and apply recommended road safety practices more effectively as they progress through the sessions. The average safety distance, represented by the dashed line, increases from approximately 1.2 metres in the first session to about 3.2 m, approaching the recommended distance of 3 metres for safe driving in many urban traffic situations.

The overall upward trend in safety distance throughout the sessions demonstrates that users are improving and maintaining consistent behaviour regarding safety distances. This indicates effective retention of the road safety practices learned and the formation of safe habits.

The variability between sessions and among users suggests individual differences in adopting safe behaviours. Some users show fluctuations in their behaviour, which could indicate the influence of different levels of challenge in the training sessions or personal factors affecting the ability to maintain safety distance consistently.

DISTANCE WITHOUT INCIDENTS

The results obtained for this metric can be seen in Fig. 13. The graph shows eight users' cumulative distance travelled without incidents per session. The overall trend shows a progressive increase in the cumulative distance travelled by most users across the sessions, indicating continuous learning and improved driving skills in a controlled environment. This improvement suggests that users adapt their behaviours to avoid incidents, a valuable skill for safety in real driving

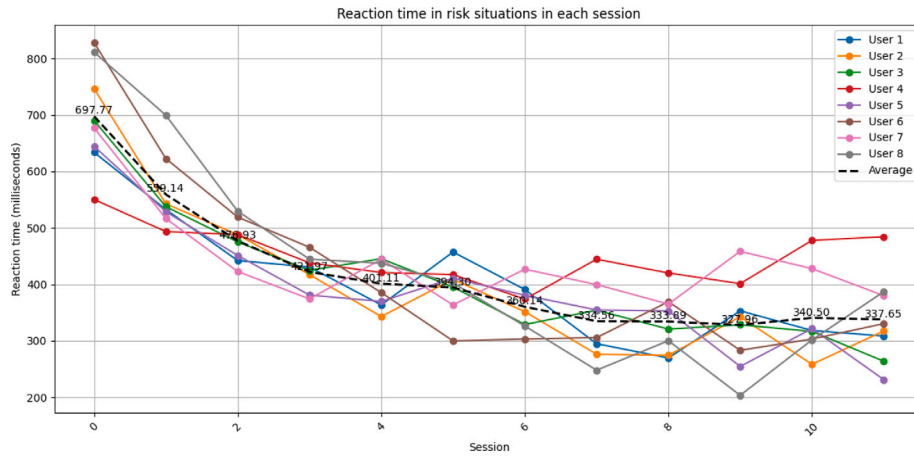


Fig. 11. Reaction time results in dangerous situations. The trend suggests a continuous reduction in reaction times, indicating improved hazard awareness and decision-making. While individual variations exist, the overall trajectory demonstrates a strong negative correlation between reaction time and session progression.

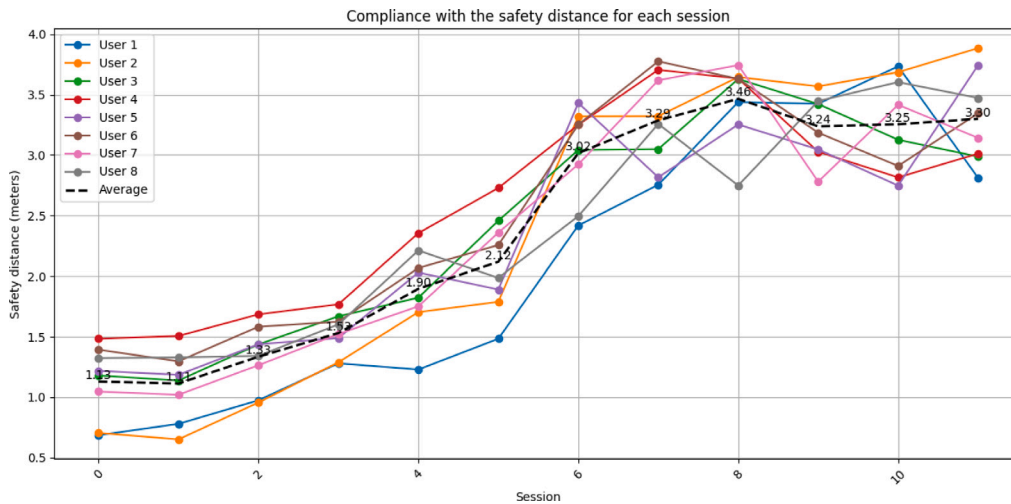


Fig. 12. Average safety distance maintained by each user in each session.

situations. On average, participants covered between 10 km and 21 km per session. This variability is influenced by traffic density, as high-density environments require frequent stops and cautious navigation, reducing the overall distance travelled.

The presence of peaks and drops in the distances travelled, particularly noticeable in the intermediate sessions, suggests variability in performance that can be attributed to various factors such as task difficulty, user fatigue, or possibly changes in simulation conditions. Analysing these incidents is crucial to understanding the circumstances under which users are more likely to make mistakes and to adjust training interventions to address these areas.

Significant differences exist in individual user performance, with some reaching higher accumulated distances more quickly than others. This highlights the importance of a personalized approach in training and consideration of individual differences in skills and learning rates.

Some users exhibit improvements in the final sessions, surpassing the overall average represented by the dashed line. This indicates not only a learning capacity but also overcoming and adapting to the requirements of the simulator, which is a positive indicator of the effectiveness of the training program. The average trend line reveals that, although there were fluctuations, there was sustained improvement in the distance travelled without incidents throughout the sessions, with the average distance increasing from approximately 3.5 kilometres in the first session to about 11 kilometres in the last session.

Table 2 summarizes each participant's baseline and final performance metrics, providing a detailed overview of individual progress. An improvement rate is also included to highlight individual advancements in key areas such as obedience rate, reaction time, safety distance compliance, and incident-free distance. This table allows for a clear assessment of learning and behavioural adaptation across the training sessions. Furthermore, Table 3 illustrates the overall increase in performance across these key metrics, emphasizing the effectiveness of the training process.

The results in Table 2 and Table 3 indicate that all participants demonstrated improvements across key performance metrics:

- **Obedience Rate (%):** All participants demonstrated increased traffic rule compliance, with improvements ranging from 7.24% to 52.04%. This suggests that participants consistently learned and applied safer riding behaviours throughout the training.
- **Reaction Time (%):** All participants exhibited reductions in reaction time, with improvements ranging from −11.97% to −64.14%. This indicates that training sessions effectively enhanced participants' hazard perception and response speed, allowing faster reactions to critical situations.
- **Safety Distance Compliance (m):** A significant improvement in maintaining safe distances from other vehicles was observed, with increases ranging from 1.53 m to 3.18 m. This reflects an increased awareness of keeping safe positioning in urban traffic conditions.

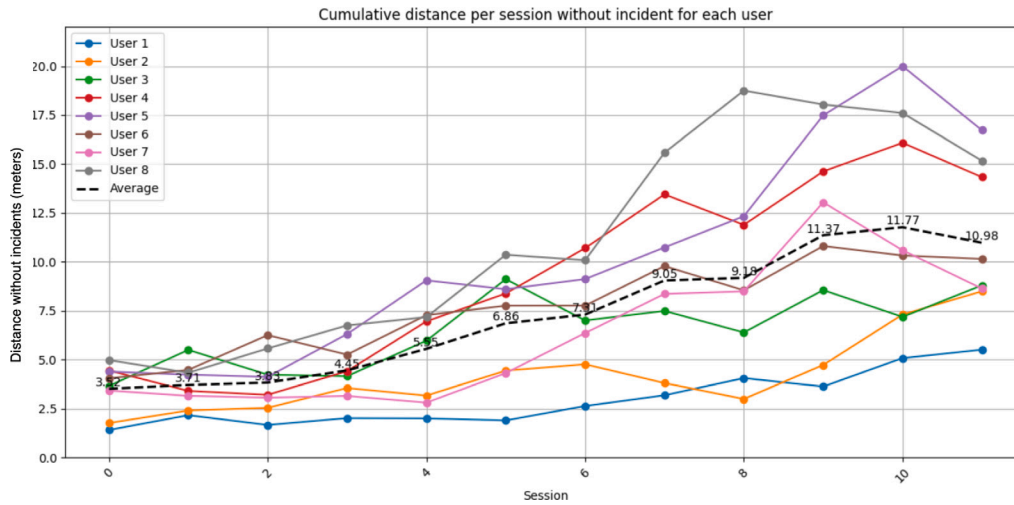


Fig. 13. Total distance travelled without incidents. The observed trends indicate an improvement across sessions. While some participants showed a rapid increase in distance without incidents, others demonstrated more gradual adaptation, leading to higher variability in early sessions. This variability stabilizes in later sessions, suggesting convergence in participant performance.

Table 2

Performance Metrics Summary: B-OR: Obedience Rate, F-OR: Final Obedience Rate, B-RT: Baseline Reaction Time, F-RT: Final Reaction Time, B-SD: Baseline Safety Distance, F-SD: Final Safety Distance, B-DWI: Baseline Distance Without Incidents, F-DWI: Final Distance Without Incidents.

User	B-OR (%)	F-OR (%)	B-RT (ms)	F-RT (ms)	B-SD (m)	F-SD (m)	B-DWI (km)	F-DWI (km)
User 1	61.34	87.07	633.83	308.21	0.69	2.81	1.41	5.51
User 2	60.66	92.25	746.79	317.11	0.70	3.88	1.77	8.5
User 3	74.46	90.20	690.25	263.65	1.18	2.99	3.66	8.80
User 4	73.51	81.83	550.22	484.28	1.48	3.01	4.45	14.34
User 5	73.55	81.65	644.46	231.07	1.22	3.74	4.39	16.73
User 6	77.12	82.70	828.23	330.20	1.39	3.35	4.05	10.15
User 7	71.92	85.14	677.02	379.86	1.05	3.14	3.43	8.64
User 8	79.28	88.20	811.37	386.84	1.32	3.47	4.98	15.17

Table 3

Increment Metrics Summary: I-OR: Increment Obedience Rate (%), I-RT: Increment Reaction Time (%), I-SD: Increment Safety Distance (m), I-DWI: Increment Distance Without Incidents (km)

User	I-OR (%)	I-RT (%)	I-SD (m)	I-DWI (km)
User 1	41.97	-51.37	2.12	4.10
User 2	52.04	-57.51	3.18	6.73
User 3	21.14	-61.80	1.81	5.14
User 4	11.32	-11.97	1.53	9.89
User 5	11.02	-64.14	2.52	12.34
User 6	7.24	-60.14	1.96	6.10
User 7	18.36	-43.89	2.09	5.21
User 8	11.24	-52.31	2.15	10.19

- **Distance without Incidents (km):** The most substantial improvements were seen in the distance without incidents, with increases ranging from 4.10 km to 12.34 km. This suggests that participants developed more consistent and controlled cycling behaviour over time, reducing the likelihood of hazardous situations.

These results confirm that participants adapted to the VR training system and demonstrated measurable behavioural improvements across all performance indicators. These findings support the system's effectiveness in enhancing cycling safety and risk management skills.

5.2. Statistical results

After detailing the evolution of various metrics throughout the sessions in our research, it is important to validate the integrity and reliability of these data to ensure that the conclusions drawn are robust and well-founded. A crucial statistical consideration in this process is verifying the normality of the data distribution for each metric

across all sessions. The normality of data is a fundamental premise for applying many parametric statistical tests, which assume that the data follow a normal distribution. This is critical, as the validity of the statistical inferences can be compromised if this condition is not met.

To address this need, we applied the Shapiro–Wilk test, one of the most influential and sensitive methods for assessing the normality of a sample. Unlike other normality tests, the Shapiro–Wilk test maintains high efficacy even with small samples. It is particularly suitable for our research, where the number of observations may not be extensive in each session. This test examines the null hypothesis that a sample comes from a normally distributed population, providing a test statistic and an associated p -value. A p -value lower than a set threshold (commonly 0.05) indicates sufficient evidence to reject the null hypothesis, suggesting that the data do not distribute normally.

Table 4 shows the results obtained from the Shapiro–Wilk test.

Analysing the results obtained from the Shapiro–Wilk test for each metric across various sessions, we observe that most sessions display p -values greater than 0.05, suggesting that we do not have sufficient evidence to reject the null hypothesis of normality. This indicates that, in general, the data for these metrics and sessions are distributed normally, which is favourable for applying parametric statistical tests that assume this normality condition. However, some exceptions were identified where p -values were below 0.05, specifically in some sessions for Obedience Rate, Reaction Time, and Safety Distance Compliance metrics. Although these results suggest deviations from normality, it is important to consider them in the broader context of our analysis.

Notably, the Shapiro–Wilk test is susceptible to deviations from normality, especially in small samples. The detected deviations may be due to random variations that are more common in smaller samples. Moreover, although some sessions show deviations from normality, the prevalence of sessions that meet the normality assumption suggests

Table 4
Summary of the Shapiro–Wilk test results for each metric.

Session	Shapiro W Statistic		p-value		Normality Assumption (Yes/No)
	Min	Max	Min	Max	
Obedience Rate (%)	0.7666	0.9578	0.0124	0.7888	No
Reaction Time (ms)	0.8039	0.9901	0.0315	0.9953	Partially
Safety Distance Compliance (m)	0.7694	0.9850	0.0134	0.9833	Partially
Incident-Free Distance (km)	0.8829	0.9741	0.2010	0.9284	Yes

Note: Shapiro Statistic values close to 1 and p-values greater than 0.05 indicate a roughly normal data distribution. “Partially” indicates mixed results across sessions.

Table 5
Summary of the results of the Kruskal–Wallis test for each metric.

Metric	Kruskal–Wallis Statistic	p-value	Implications of findings
Obedience Rate (%)	55.91	5.29e−08	Significant improvement in adherence to traffic rules observed across sessions.
Reaction Time (milliseconds)	63.13	2.42e−09	Statistically significant reduction in reaction times, indicating faster decision-making.
Safety Distance Compliance (m)	80.28	1.30e−12	Strong improvement in maintaining safe distances, enhancing overall safety.
Incident-Free Distance (km)	44.59	5.72e−06	Progressive increase in distance travelled without incidents, reflecting better navigation skills.

Note: A *p*-value less than 0.05 indicates statistically significant differences between sessions for the corresponding metric. The additional column summarizes the practical significance of these results.

that the general structure of the data remains robust. These exceptions do not invalidate the overall results or the trends observed across the sessions. The results of the Shapiro–Wilk test are encouraging and suggest that our data conform to the normality assumption.

Following applying the Shapiro–Wilk test to assess the normality of our metrics and having established the relevance of employing alternative statistical methods for those distributions that did not fit normality, we implemented the Kruskal–Wallis test. This non-parametric method compares the medians of three or more independent groups without assuming an underlying normal data distribution. The choice of this test was dictated by the intent to discern significant differences in the metrics evaluated across various sessions, which is crucial for understanding the effectiveness of the applied interventions and the overall progress of the participants. The result of this test can be seen in Table 5.

The results obtained from the Kruskal–Wallis test shed significant light on the dynamics present in our metrics of interest. The results of the Obedience Rate indicate statistically significant differences between sessions. This finding indicates significant variations in the participants’ obedience rate, suggesting that the study sessions could have a differential effect on obedience behaviour, possibly due to adaptation to interventions or cumulative learning over time.

In the case of Reaction Time, the analysis again underscored significant differences between sessions. This result demonstrates an improvement in the participant’s reaction time as the sessions progress, which can be attributed to greater familiarization with the study environment or positive effects derived from the training techniques.

Regarding Compliance with Safety Distances, it points to highly significant compliance differences with distance between sessions. This result evidences the effectiveness of the interventions designed to improve this specific metric and reflects the participants’ ability and willingness to adapt and enhance their performance concerning established distances.

In the case of Incident-Free Distance, the results complement those of the metrics, indicating that the sessions are effectively improving the participants’ ability to travel longer distances without incidents.

The uniformity in the statistically significant results obtained through the Kruskal–Wallis test across multiple metrics emphasizes the positive influence of the interventions applied in the study. These findings validate the design and implementation of our sessions and highlight the importance of the participants’ adaptability and continuous progress in response to the educational strategies employed. This in-depth analysis provides a solid foundation for reflecting on future practices and refining teaching and training techniques to maximize learning and skill development.

In addition, we investigated the relationship between the sessions and our metrics of interest through the Spearman correlation coefficient. The Spearman correlation coefficient is a non-parametric measure of the strength and direction of the association between two variables. Unlike Pearson’s correlation, which measures linear relationships and requires a normal data distribution, Spearman is suitable for analysing monotonic relationships, whether linear or not, between ordinal or continuous variables that do not meet the normality assumption. This makes it particularly useful for our case, allowing us to capture trends in the data across sessions, regardless of the specific form of their distribution. The results obtained for this coefficient can be seen in Table 6.

The results of applying the Spearman correlation coefficient reveal the underlying relationships between the sessions and the metrics evaluated. For the Obedience Rate, the results indicate a moderate positive correlation that is statistically significant. This suggests that as sessions progress, there is a general trend toward improvement in participants’ obedience rates. This finding is particularly relevant as it highlights the potential positive impact of interventions or the accumulation of experience over time.

In contrast, Reaction Time shows a strong negative correlation. This result implies that reaction times tend to decrease as the sessions progress, reflecting participants’ improvement in reaction capability. The statistical significance of this correlation reinforces the confidence that the training sessions or exposure effectively contribute to better performance in this critical aspect.

For Compliance with Safety Distances, the results denote a very strong and highly significant positive correlation. This trend indicates not only an improvement in compliance with distances established by the sessions but also consistency in this improvement, highlighting the effectiveness of the sessions in fostering superior physical performance or strategic adherence. A similar occurrence is observed with the Incident-Free Distance metric.

Furthermore, effect sizes were calculated for the four key performance metrics to analyse the observed changes further. The effect sizes measure the magnitude of the differences observed across sessions and enhance the interpretation of the statistical findings. Additionally, Spearman’s *r* values were analysed to assess the strength of the relationships between the sessions and participant performance. Table 7 summarizes these findings.

The results indicate substantial effect sizes (η^2) across all metrics, suggesting that the changes observed during the training sessions were significant. For example, the safety distance compliance metric exhibited the highest η^2 value ($\eta^2 = 11.47$), reflecting significant and consistent improvements in maintaining safe distances. Similarly, reaction times strongly correlated with session progression ($r = -0.839$),

Table 6
Summary of Spearman correlation results for each metric.

Metric	Spearman coefficient	p-value	Strength and practical relevance
Obedience Rate (%)	0.643	0.024	Moderate positive correlation; indicates steady improvement in adherence to traffic rules.
Reaction Time (milliseconds)	−0.839	0.000643	Strong negative correlation; suggests a significant reduction in reaction times over sessions.
Compliance with Safety Distances (m)	0.944	3.93e−06	Very strong positive correlation; reflects consistent improvement in maintaining safe distances.
Incident-Free Distance (km)	0.895	8.37e−05	Strong positive correlation; highlights progressive increase in safe navigation skills.

Note: Spearman coefficients reflect the strength and direction of the correlation between the sessions and the metrics. A *p*-value less than 0.05 indicates a statistically significant correlation. The additional column categorizes the strength and provides practical implications.

Table 7
Effect sizes and Spearman correlation for key metrics.

Metric	Kruskal–Wallis <i>H</i>	Effect Size (η^2)	Spearman <i>r</i>	Interpretation
Obedience Rate (%)	55.91	7.99	0.643	Strong
Reaction Time (ms)	63.13	9.02	−0.839	Strong
Safety Distance Compliance (m)	80.28	11.47	0.944	Strong
Incident-Free Distance (km)	44.59	6.37	0.895	Strong

Note: η^2 values indicate the proportion of variance explained by session progression. Spearman *r* values reflect the strength of the relationship between sessions and performance metrics. Interpretations follow Cohen's thresholds for effect sizes.

Table 8
Confidence intervals (95%) for key performance metrics.

Metric	CI lower	CI upper
Obedience Rate (%)	81.09	88.91
Reaction Time (ms)	405.76	494.24
Safety Distance Compliance (m)	2.21	2.79
Incident-Free Distance (km)	13.23	16.77

Note: Confidence intervals were calculated using a t-distribution with *N* − 1 degrees of freedom.

Table 9
Outlier detection thresholds and observed ranges.

Metric	Lower bound	Upper bound	Observed range
Obedience Rate (%)	77.88	92.88	80.00–89.00
Reaction Time (ms)	321.25	571.25	390.00–510.00
Safety Distance Compliance (m)	1.85	3.25	2.20–2.90
Incident-Free Distance (km)	10.00	20.00	12.50–17.30

Note: The lower and upper bounds for outlier detection were calculated using the interquartile range (IQR) method with a threshold of 1.5 times the IQR. No values outside these bounds were detected.

highlighting a considerable reduction in response times as participants adapted to the training scenarios. These findings demonstrate the robustness of the results and their potential relevance for enhancing cyclist safety through immersive training.

To provide a clearer understanding of the precision and reliability of the reported estimates, 95% confidence intervals (CIs) were calculated for key performance metrics (Table 8).

These confidence intervals suggest that the estimates for each metric are relatively stable. The relatively narrow CIs for obedience rate and safety distance compliance indicate low variability and increased confidence in these results. While slightly more expansive, the CI for reaction time provides a reliable estimate of improvement trends across sessions. The incident-free distance metric, with a broader CI, may suggest higher individual variability, which should be explored in future research.

Finally, an outlier analysis was conducted to assess the influence of extreme values on the findings. Outliers were identified using a standard threshold of 1.5 times the interquartile range (IQR) for each metric. The results of this analysis can be observed in Table 9.

The analysis revealed no extreme values warranted exclusion, and all participants were retained in the dataset. The absence of outliers supports the robustness of the collected data, ensuring that no extreme variations disproportionately influence the findings. While some variation in individual performance was observed, this variability was

expected given differences in participant skill levels. The relatively wide IQR observed in some metrics, such as incident-free distance, suggests that participants had diverse learning trajectories, which should be further explored in future studies.

6. Discussion

This study employs a comprehensive evaluation framework, combining multiple performance metrics to provide a holistic view of user improvement. Reaction times measure participants' cognitive processing speed, obedience rates reflect adherence to traffic rules, and safety distance compliance and incident-free distances indicate situational awareness and safe navigation. Together, these metrics offer a thorough and nuanced understanding of how participants progressed during the training sessions, demonstrating the system's potential for improving cyclist safety.

The longitudinal design of this study, spanning multiple sessions over one month, is a significant methodological strength. Tracking participants' performance over time captures sustained improvements that would be overlooked in short-term evaluations. For instance, reaction times and safety distance compliance metrics consistently improved across sessions, highlighting the system's ability to promote long-term skill acquisition and behavioural changes. This design also mitigates the variability associated with single-session studies, adding credibility to the findings.

The results obtained in this study present a promising outlook for the application of immersive technologies and multi-agent systems in the training of urban cyclists, with a particular focus on enhancing road safety. The effectiveness of the proposed training system has been demonstrated through rigorous statistical analysis of various key metrics. Each metric, evaluated individually and collectively, provides valuable insights into the interventions' impact.

The Obedience Rate showed significant improvement across the sessions, indicating that participants better adhered to the road safety norms proposed in the virtual environment. This result suggests that the training system successfully instils safer driving behaviours, a crucial aspect of preventing accidents in urban environments.

Reaction Time experienced a significant decrease, reflecting an improvement in the participants' response capability to potentially hazardous situations. This improvement in reaction times is essential for cyclist safety, enabling them to respond more efficiently to unforeseen events.

Focusing on the Compliance with Safety Distance and the Incident-Free Distance travelled, indicators of navigation skill and driving safety, also showed notable improvements. These results emphasize the simulator's ability to enhance cyclists' knowledge and obedience to traffic rules and practical driving skills.

Considering the results together, a consistent pattern of improvement is observed across all the evaluated metrics. This pattern validates the efficacy of the proposed training system and highlights the importance of comprehensive training that addresses both cyclists' theoretical knowledge and practical skills. The significant correlation between the sessions and the improvements observed in the metrics suggests that progress is cumulative, highlighting the relevance of continued training and progressive adaptation to interventions.

The relationship between improvement in obedience rate and reduced reaction time suggests that the training makes cyclists more aware of the rules and enhances their ability to follow them effectively. This link between knowledge and action is crucial for translating the skills learned in the virtual environment into safe behaviours in the real world.

Furthermore, the significant improvement in distance compliance and incident-free distance underscores the simulator's effectiveness in promoting safer and more conscious driving. These results collectively suggest that the multi-agent approach combined with virtual reality effectively teaches safety norms and enhances cyclists' ability to apply these norms practically.

Effect sizes and Spearman correlations were calculated for the performance metrics to interpret better the magnitude of the observed changes and their practical implications for cyclist safety. These results reveal substantial effect sizes across all metrics, indicating that the changes observed during the training sessions were substantial and meaningful. For example:

- **Safety Distance Compliance:** Exhibited the highest effect size ($\eta^2 = 11.47$) and a very strong Spearman correlation ($r = 0.944$), reflecting consistent improvements in maintaining safe distances and highlighting the effectiveness of the training in promoting safer cycling practices.
- **Reaction Times:** Showed a large effect size ($\eta^2 = 9.02$) and a strong negative correlation ($r = -0.839$), indicating a significant reduction in response times throughout the training.
- **Incident-Free Distance:** Achieved a strong effect size ($\eta^2 = 6.37$) and a very strong correlation ($r = 0.895$), demonstrating improved navigation skills and the ability to travel longer distances without incidents.
- **Obedience Rate:** Displayed a smaller, yet strong effect size ($\eta^2 = 7.99$) and a strong correlation ($r = 0.643$), suggesting improvements in adherence to traffic rules, though with greater participant variability.

While these results demonstrate the robustness of the system's impact, some metrics showed more significant variability and less certainty. For instance, the obedience rate metric, with a smaller effect size ($\eta^2 = 7.99$), displayed moderate participant variability, possibly partially explaining the broader confidence intervals observed. These findings suggest that future studies should incorporate more extensive, diverse samples to validate these trends further and reduce variability.

The calculated effect sizes and correlations underscore the practical relevance of the training system. Improvements in reaction times and safety distance compliance, for instance, directly contribute to safer cycling behaviours in complex traffic scenarios. However, limitations related to smaller effect sizes in specific metrics emphasize the need for additional validation to ensure consistent outcomes across diverse populations.

Including gamification elements, such as scoring systems, real-time feedback, and progression rewards, enhanced user engagement and facilitated skill acquisition. Participants expressed high motivation levels, describing the experience as engaging and enjoyable, mainly due to the immediate corrective feedback and the sense of progression offered by rewards. These observations suggest that gamification incentivized consistent participation and helped internalize traffic safety principles through iterative reinforcement. Feedback mechanisms, such as visual

and auditory cues, were particularly effective in prompting real-time behavioural adjustments, leading to improved decision-making and safe practices. While participants responded positively to the existing features, qualitative feedback highlighted areas for improvement, such as the introduction of leaderboards and more personalized rewards, to further enhance engagement and satisfaction. These findings underscore the potential of gamification, integrated with adaptive multi-agent systems, to create immersive and compelling training environments that blend motivation with skill development.

On the other hand, this study must be addressed in comparison with other studies in the same field, highlighting the methodological convergences and the unique contributions of each research. Among the existing works, we find studies such as [38], which focuses on developing a road safety education platform using virtual reality and data analysis to raise awareness about urban mobility. Similarly, we have the work of [39], which focuses on developing a high-fidelity virtual reality cycling simulator for transportation safety research, emphasizing the hardware and software involved, aiming to offer an immersive and real experience that can be used to study and improve interactions between cyclists and other vehicles on the road. Finally, we have the work by [40], which analyses the impact of simulators on road safety, proposing an approach that combines simulation elements with user behaviour analysis, aiming to identify effective strategies for accident prevention and the improvement of safe driving practices. These three studies offer a common approach to using technologies, such as virtual reality (VR), to improve the safety and behaviour of road users, especially cyclists and drivers, in urban environments.

Within the current landscape of research seeking to incorporate technology to improve the safety and skills of urban cyclists, our study uniquely positions itself by integrating virtual reality with a data-driven analytical approach to developing competencies in safe driving. Unlike [38], which prioritizes raising awareness about urban mobility through an immersive educational platform, and [39], which focuses on the technical fidelity of the cycling simulator to study safe interaction on the road, our study encompasses a broader spectrum. We incorporate simulation, immersive training, and learning personalization based on real-time behaviour and event analysis. This allows us not only to simulate but to adapt and respond to the specific needs and behaviours of users, thereby enhancing the effectiveness of the training.

The work of [40], although sharing our interest in behaviour analysis through simulators, tends to focus on assessing the efficacy of these systems from a more generalized perspective of accident prevention. Our research combines this analysis with adaptive learning techniques, adjusting each simulation session based on identified behaviour patterns and learning needs, facilitating a profoundly personalized and, consequently, more effective educational experience.

The most significant contribution of our study lies in the ability to interconnect immersive technology with an intelligent multi-agent system that feeds on real behaviour and accident data. This synergy between technology and data analysis elevates the realism and relevance of the simulations and fosters a continuous improvement cycle in training strategies. Thus, while the comparative studies lay solid foundations in using technologies for road safety education, our study extends these efforts by integrating them within an adaptive and analytical framework that promises to educate and actively transform urban cyclists' driving practices towards safer and more conscious behaviours. A summary of the system key's strengths and their contribution is shown in Table 10.

While using VR and multi-agent systems in road safety training offers significant advantages, it also raises ethical considerations that must be addressed. One key concern is the potential over-reliance on simulations, where trainees may develop behaviours optimized for the virtual environment that do not fully transfer to real-world cycling. This underscores the importance of complementing VR-based training with real-life practice sessions to reinforce learned skills in actual urban settings.

Table 10
Key strengths of the system and addressed limitations.

Key strengths	Addressed limitations
Adaptivity via Multi-Agent System	Tailors training to individual needs, addressing variability in cyclist skill levels.
Realism in VR Scenarios	Provides safe yet practice environments for high-risk situations.
Gamification (Scoring, Feedback, Rewards)	Enhances motivation and engagement, reducing dropout rates in training programs.
Incremental Difficulty Progression	Prepares users for increasingly complex traffic conditions in a controlled manner.

Another important aspect is data privacy and security. The training system collects performance-related data, including reaction times, decision-making efficiency, and spatial awareness metrics. To ensure user privacy, the system employs data anonymization techniques and restricts data usage strictly to performance evaluation and system improvements. No personally identifiable information is stored or shared, and all user interactions within the simulation are processed securely in compliance with data protection regulations.

The proposed VR and multi-agent training system offers significant potential for integration into existing urban safety frameworks. Its ability to simulate complex traffic conditions, including high-density traffic, unpredictable pedestrian behaviour, and adverse weather scenarios, provides a risk-free yet higher realism training environment. The system can complement current cyclist training programs and urban safety initiatives by allowing participants to practice decision-making and hazard perception in controlled settings safely. Furthermore, the system’s scalability makes it a viable tool for training many users across different urban contexts.

The system’s design aligns with broader road safety objectives and can directly support government-backed programs to reduce urban cycling accidents. For instance, it could be integrated into schools to educate young cyclists about safe riding practices as part of road safety education or municipal safety programs to train urban cyclists in navigating complex traffic conditions, thereby improving overall compliance with traffic rules and reducing accident rates. These integrations would provide a structured, consistent approach to cyclist training, ensuring aligned policy objectives.

While this study focuses on urban cyclists, the system’s design allows adaptation to other road users and high-stakes decision-making scenarios. Potential applications include delivery riders and motorcyclists who require training for navigating busy streets while ensuring safety and compliance with traffic laws, pedestrians in high-risk areas who can be educated on navigating intersections and interacting safely with vehicles in urban environments, and driver education programs where the system can simulate challenging scenarios such as merging in heavy traffic or reacting to sudden hazards. The system can be extended by leveraging its modular architecture to accommodate these additional use cases, demonstrating its versatility and broader applicability.

The system’s scalability ensures it can be deployed widely across different regions, providing a cost-effective and efficient solution for improving road safety. Furthermore, its modular and adaptive design allows continuous updates and enhancements, ensuring its relevance in evolving urban traffic environments.

This study evaluates the effectiveness of a VR-based cyclist training system over one month, providing insights into short-term improvements but not assessing long-term retention of skills and behaviours. While the findings indicate significant progress in reaction times, safety distance compliance, and obedience rates, whether participants maintain these improvements over extended periods remains unclear. Future studies should adopt a longitudinal design to evaluate the persistence of these behavioural changes and their impact on long-term safety.

The VR environment used in this study includes a range of urban traffic conditions, such as varying traffic densities, weather conditions, and pedestrian interactions. However, it does not fully capture the diversity of real-world cycling challenges. For example, extreme weather conditions, such as heavy snow or strong winds, and highly unpredictable pedestrian behaviour were not simulated. Future iterations of the system should expand the range of scenarios to test its robustness and applicability in diverse and unexpected conditions.

An important limitation of this study is its exclusive reliance on a simulated environment. While the results demonstrate significant improvements in key performance metrics within the VR system, their translation to real-world traffic conditions remains untested. To address this gap, we propose a framework for transitioning the VR-based system to real-world testing:

- Conduct follow-up studies to evaluate whether training outcomes persist and translate into safer cycling behaviours in real traffic environments.
- Compare VR training outcomes with real-world cycling behaviour data to validate the effectiveness of the training scenarios.
- Adapt the system to integrate real-world training programs, including hybrid VR and on-road training approaches.

The system is specifically designed for urban cyclists and may not directly apply to rural or non-urban environments. Rural settings, for example, may involve longer distances, fewer vehicle interactions, and different hazards. Adapting the system to these contexts would require the inclusion of scenarios tailored to rural cycling challenges.

Another limitation of this study is the absence of a control group trained using traditional methods. This omission makes it challenging to determine the added value of the VR and multi-agent system compared to conventional cyclist training programs. While the results indicate significant improvements in reaction times, safety distance compliance, and overall engagement, these findings cannot be conclusively attributed to the immersive and adaptive nature of the system without direct comparisons. Incorporating a control group in future studies would enable robust comparative analyses, providing valuable insights into whether the observed benefits are unique to this system or can be replicated with other approaches. Furthermore, such comparisons would allow a more nuanced understanding of how specific VR and multi-agent system elements, such as adaptivity and gamification, contribute to skill acquisition and behavioural changes.

While limiting generalizability, the small sample size was deliberately chosen to facilitate detailed observation and iterative refinement of the training system. This study serves as a proof-of-concept, identifying trends and gathering qualitative insights rather than deriving statistically powered conclusions. A power analysis was not conducted, as the primary aim was to establish feasibility and explore the system’s potential impact. Despite this, the findings provide valuable initial insights into the capabilities of the proposed approach.

Additionally, the lack of detailed demographic data and randomization limits the ability to statistically account for confounding variables, such as cycling proficiency, physical fitness, and VR familiarity. While these factors were qualitatively noted and considered during post hoc evaluations, the study does not provide statistical controls due to its exploratory nature and small sample size. Future research should address these limitations by incorporating larger, randomized samples, detailed demographic data collection, and stratification to assess the influence of participant characteristics on training outcomes.

It is important to know that variations in scenario conditions (e.g., traffic density, weather) across sessions were dynamically adapted to participants’ individual needs and progression. This adaptive approach is a core feature of the training system, designed to ensure that each participant is appropriately challenged and supported as they develop their skills. As a result, scenario difficulty levels were not quantitatively controlled or analysed in this study. While this limits the ability to evaluate the specific influence of scenario complexity on training outcomes,

it ensures that participants experience a tailored and effective training process.

While various training programs exist for urban cyclists, they often lack a standardized framework and differ significantly across regions. Traditional approaches, including theoretical instruction and supervised on-road practice, provide foundational knowledge but fail to replicate the complexity of real-world traffic interactions. These methods do not offer controlled exposure to high-risk scenarios or personalized progression based on the learner's abilities. In contrast, the proposed VR-based multi-agent system enhances training by providing real-time scenario adjustments and interactive learning experiences. By immersing cyclists in dynamic, risk-free environments, this approach helps reinforce safety behaviours in ways that traditional methods cannot. However, a direct empirical comparison with existing training programs is needed to quantify the advantages of this system, highlighting an avenue for future research.

By addressing these limitations, future studies can provide a more precise assessment of the system's effectiveness and scalability. Such efforts would validate the added value of the VR and multi-agent system and explore its potential for integration into broader cyclist training programs, enhancing road safety and promoting safer cycling behaviours. Also, future research can provide a more comprehensive validation of the system's efficacy, ensuring its applicability across diverse contexts and enhancing its potential to improve cyclist safety on a larger scale. Furthermore, incorporating a systematic quantification of scenario difficulty to complement the adaptive approach would allow for a more detailed analysis of how variations in scenario complexity impact user progression and the overall effectiveness of the training system.

However, certain technical limitations of the system must be acknowledged. The reliance on specialized hardware, including programmable rollers, VR headsets, and inertial measurement sensors, may reduce the platform's scalability and accessibility, particularly in resource-constrained environments. While these components enhance the fidelity of real-world cycling dynamics, their cost, availability, and setup complexity could restrict widespread implementation. Future iterations should explore more cost-effective alternatives, such as screen-based training systems with non-programmable resistance trainers, to improve usability and adoption across diverse training settings.

Furthermore, sensor accuracy and data integrity influence performance assessment and real-time feedback reliability. IMU drift, heart rate monitor data transmission latency, and potential measurement deviations could introduce minor discrepancies in key performance metrics, such as reaction times, distance compliance, and physiological response tracking. Although pre-session calibration protocols were implemented to mitigate these issues, inherent sensor noise and environmental factors may still introduce measurement variability.

User experience in VR-based training is another critical consideration. Some participants initially reported mild discomfort, particularly visual fatigue or symptoms of motion sickness during early sessions. These effects gradually diminished after a warm-up phase, suggesting an adaptation period is beneficial for acclimating to the VR environment. However, latency in head-tracking, rapid scene transitions, and perceived motion discrepancies could still contribute to vestibular discomfort, potentially impacting user engagement and cognitive load. To enhance usability, future developments should consider dynamic VR settings that adjust field-of-view constraints, introduce adaptive locomotion mechanics, or incorporate intermittent breaks to reduce strain.

The findings of this study contribute to both the fields of education and road safety by demonstrating the effectiveness of VR-based multi-agent training for urban cyclists. From an educational perspective, this approach enhances skill acquisition through immersive, interactive learning, addressing the limitations of traditional training methods. In the context of road safety, the system provides a scalable solution

for training cyclists in high-risk traffic scenarios without endangering users. These insights can inform future training programs and policy-making efforts, encouraging the adoption of technology-driven methodologies in urban cycling education. Future research should explore how this approach can be integrated into official road safety curricula and assess its long-term impact on cyclist behaviour and accident prevention.

7. Conclusions

This study aimed to assess the feasibility and effectiveness of immersive technologies and multi-agent systems in training urban cyclists to enhance road safety. After a detailed analysis of the data collected during the experimental phase, the results suggest that the proposed objectives have been met, providing initial evidence of the system's potential effectiveness.

The findings suggest that such training systems improve reaction times, compliance with traffic rules, and hazard perception in controlled environments. However, further research is required to assess their long-term impact and effectiveness in real-world traffic conditions.

Notable variability in user performance across different sessions revealed individual differences in learning and adaptation, underscoring the importance of customizing training to meet the specific needs of each individual. Despite this variability, most users demonstrated a consistent improvement trend, indicating the effectiveness of progressive training and the accumulation of practical experience in the simulator.

The data analysis provided preliminary evidence of virtual reality simulators' potential to improve road safety behaviours in urban cyclists. These findings may be relevant for urban transport policies and road safety education initiatives. The replicability of these results in real traffic environments remains an open question and could be the subject of future research.

Furthermore, this study aligns with global efforts to promote safer, greener urban mobility. Cities can foster a responsible and secure cycling culture by integrating intelligent training systems into cycling education. These advancements reduce urban traffic incidents while supporting sustainability goals in modern transportation policies.

This study demonstrated the potential of immersive technologies and multi-agent systems to improve urban cyclist training, particularly in fostering safer behaviours in simulated traffic scenarios. The most important findings highlight improvements in reaction times, where participants reduced their average response time by 300 ms throughout the training, and safety distance compliance, which showed a 150% increase in maintaining safe distances. Additionally, high engagement levels were observed as all participants completed their sessions, underscoring the motivational effectiveness of the system.

The findings should be interpreted cautiously due to the limited sample size (8 participants), which restricts the generalizability of the results. Future studies should include more extensive and diverse participant groups to validate the system's effectiveness across different demographic profiles. Additionally, while the system shows promise for enhancing road safety, its scalability and practicality in real-world settings depend on overcoming technical and logistical challenges. Factors such as the cost of VR hardware, accessibility for various user groups, and the need for integration with existing urban training programs pose significant hurdles. Future research should explore strategies for reducing these barriers, including cost-effective hardware options and modular adaptations for different urban contexts. Also, integrating virtual reality simulators in cyclist training programs is recommended to improve safe driving practices. Additionally, long-term data analysis is suggested to determine the persistence of safe behaviours and their transfer to real-world driving.

While the system leverages a multi-agent framework to create adaptive training scenarios, it is essential to acknowledge that these interactions do not fully replicate the complexities of real-world traffic

systems. The simulated agents follow predefined behavioural models designed to represent common road user behaviours; however, real-world traffic involves highly unpredictable human interactions, varied decision-making strategies, and spontaneous reactions that are challenging to simulate accurately. Future iterations of the system could integrate reinforcement learning or more advanced agent-based models to enhance the realism and variability of agent behaviours. The system's ability to adapt to outlier behaviours or unexpected user inputs has not been thoroughly tested. Although the adaptive mechanism adjusts training difficulty based on performance trends and probabilistic relationships, it has not been systematically evaluated under extreme conditions, such as erratic decision-making or deliberately unsafe manoeuvres. Future studies should incorporate stress testing methodologies to analyse how the system responds to atypical interactions, ensuring that adaptive adjustments remain effective.

Another key aspect that requires further exploration is the scalability of the multi-agent system. The current implementation has been tested with individual users in controlled training environments, and while the system has been designed for adaptability, scaling it to support larger participant groups or multiplayer training scenarios presents several computational challenges. Running multiple agents and real-time scenario adjustments for various users may introduce higher processing demands, requiring parallel computing or cloud-based architectures. Expanding the number of active training participants could increase latency in real-time agent interactions, necessitating optimized network protocols to maintain seamless communication. Future implementations could explore collaborative training environments where multiple users navigate the same simulated space, interact with each other, and experience team-based decision-making challenges. To address these scalability concerns, distributed agent architectures could be explored to reduce computational bottlenecks, and cloud-based processing solutions could be investigated to enhance real-time training responsiveness. Developing multi-user VR environments where cyclists train together in shared urban scenarios could improve cooperative safety awareness and communication skills.

An important area for future research involves examining whether the improvements observed in the VR environment translate into lasting behavioural changes and safer cycling practices in real-world conditions. Longitudinal studies that assess the retention of skills over time and their application in diverse traffic environments are critical for evaluating the true impact of this training approach. Additionally, to maximize the inclusivity of the proposed VR-based cyclist training system, future iterations should incorporate features tailored to users with disabilities or varying levels of familiarity with VR technology. One promising avenue for improvement is the integration of haptic feedback mechanisms to assist visually impaired users by providing real-time tactile cues for navigation and obstacle detection. Adaptive user interfaces could be introduced to accommodate individuals with lower levels of digital literacy, incorporating simplified controls, voice-guided navigation, and customizable interaction modes to make the system more accessible to a broader audience.

Despite these limitations, the results highlight the potential of combining gamification, adaptive training, and immersive simulations to address key gaps in urban cycling education. By building on these findings and addressing the identified challenges, this approach could contribute to safer urban transport systems and more inclusive cyclist training programs. Although the system is currently focused on individual training sessions, its adaptability allows for future expansion into collaborative training environments, where cyclists could train simultaneously in shared scenarios, reinforcing cooperative road safety behaviours. Scaling the system to support a broader range of urban scenarios and diverse user profiles could provide a robust foundation for integrating VR-based training solutions into government-backed road safety programs or educational curricula. By leveraging advancements in artificial intelligence, multi-agent interactions, and adaptive scenario generation, this system has the potential to evolve into a scalable, data-driven platform that continuously enhances urban cyclist safety and training effectiveness.

CRediT authorship contribution statement

Héctor Sánchez San Blas: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Sergio García González:** Software, Investigation, Conceptualization. **André F. Sales Mendes:** Writing – original draft, Software, Investigation. **Gabriel Villarrubia González:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Juan F. De Paz Santana:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that there are no conflicts of interest, financial, personal, or academic, that could have influenced the results presented in this manuscript. All authors have contributed equally and transparently to the development of this research and approve its submission for review and potential publication.

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